
VELLORE INSTITUTE OF TECHNOLOGY
CHENNAI CAMPUS

DATABASE SYSTEMS PROJECT
INITIAL REVIEW REPORT
FALL SEMESTER 2025–26

*"Informed in an Instant:
World In Brief — An AI-Driven, Database-Backed News Intelligence
Platform"*

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REVIEW STAGE	INITIAL PROJECT REVIEW

1. INTRODUCTION

WORLD IN BRIEF IS AN AI-POWERED NEWS INTELLIGENCE PLATFORM DESIGNED TO HELP USERS NAVIGATE TODAY'S OVERWHELMING VOLUME OF DAILY NEWS. THE SYSTEM AGGREGATES ARTICLES FROM MULTIPLE SOURCES INTO A SINGLE, STRUCTURED DATABASE AND LAYERS AN ON-DEVICE AI AGENT ON TOP OF IT, ALLOWING USERS TO RECEIVE SUMMARIZED, PERSONALIZED, AND INTELLIGENTLY RANKED NEWS CONTENT INSTEAD OF SCROLLING THROUGH DISCONNECTED HEADLINES.

AT ITS CORE, THE PLATFORM IS A RELATIONAL-DATABASE-DRIVEN APPLICATION: EVERY ARTICLE, USER PREFERENCE, INTERACTION, AND NOTIFICATION IS MODELLED AS A NORMALIZED ENTITY, ENSURING THAT PERSONALIZATION, SEARCH, AND ANALYTICS FEATURES CAN BE BUILT ON A CONSISTENT AND RELIABLE DATA FOUNDATION RATHER THAN AD-HOC STORAGE.

1.1 PROBLEM STATEMENT

READERS TODAY ARE EXPOSED TO A CONSTANT, UNFILTERED STREAM OF NEWS ACROSS COUNTLESS SOURCES, MAKING IT DIFFICULT TO IDENTIFY WHAT IS RELEVANT, TRUSTWORTHY, AND CONCISE. EXISTING NEWS AGGREGATORS LARGELY PRESENT RAW ARTICLES WITHOUT MEANINGFUL PERSONALIZATION, SUMMARIZATION, OR INTELLIGENT CATEGORIZATION, LEAVING USERS TO MANUALLY FILTER THROUGH NOISE TO FIND THE INFORMATION THAT MATTERS TO THEM.

1.2 OBJECTIVES

- DESIGN A NORMALIZED RELATIONAL DATABASE CAPABLE OF SUPPORTING AUTHENTICATION, PERSONALIZATION, AND ANALYTICS FEATURES AT SCALE.
- BUILD A SECURE AUTHENTICATION AND ROLE-BASED ACCESS MODULE FOR END USERS AND ADMINISTRATORS.
- INTEGRATE A LOCAL AI AGENT (LLAMA 3.2 VIA OLLAMA) TO SUMMARIZE, CLASSIFY, TRANSLATE, AND EXPLAIN NEWS ARTICLES ON DEMAND.
- DELIVER A PERSONALIZED NEWS FEED DRIVEN BY STORED USER PREFERENCES AND READING HISTORY.
- PROVIDE TRENDING TOPICS, BOOKMARKING, AND NOTIFICATION FEATURES BACKED BY EFFICIENT, INDEXED DATABASE QUERIES.
- ENSURE THE SYSTEM IS TESTABLE, SECURE, AND DEPLOYABLE WITHIN A CONTAINERIZED, PRODUCTION-READY ENVIRONMENT.

2. PROJECT SCOPE & MODULES

THE SYSTEM SCOPE IS ORGANIZED INTO SEVEN FUNCTIONAL MODULES, EACH MAPPED DIRECTLY TO A SET OF RELATIONAL DATABASE ENTITIES AND A CORRESPONDING PHASE OF THE PROJECT'S WORK BREAKDOWN STRUCTURE. TOGETHER THESE MODULES COVER 100% OF THE PROJECT SCOPE, FROM USER-FACING FEATURES DOWN TO ADMINISTRATIVE CONTROLS.

MODULE	DESCRIPTION
AUTHENTICATION	USER REGISTRATION, LOGIN, JWT-BASED SESSIONS, AND PASSWORD RESET.
PERSONALIZED NEWS	FAVORITES, READING HISTORY, AND PREFERENCE-DRIVEN FEED RANKING.
AI INTELLIGENCE	SUMMARIZATION, CLASSIFICATION, SENTIMENT ANALYSIS, AND TRANSLATION VIA A LOCAL LLM.
TRENDING & DISCOVERY	COMPUTATION OF TRENDING TOPICS AND POPULAR-ARTICLE SURFACING.
NOTIFICATIONS	BREAKING-NEWS AND DIGEST ALERTS BASED ON STORED USER PREFERENCES.
BOOKMARKS	SAVING, ORGANIZING, AND OFFLINE ACCESS TO ARTICLES.
FEEDBACK & ADMIN	USER FEEDBACK, BUG REPORTING, AND AN ADMINISTRATIVE CONSOLE.

3. DATABASE DESIGN OVERVIEW

THE DATABASE IS THE STRUCTURAL BACKBONE OF WORLD IN BRIEF. IT IS DESIGNED AROUND A 12-ENTITY RELATIONAL SCHEMA, NORMALIZED TO THIRD NORMAL FORM (3NF) TO ELIMINATE REDUNDANCY AND MAINTAIN DATA INTEGRITY ACROSS THE AUTHENTICATION, PERSONALIZATION, AND AI-INTELLIGENCE MODULES. CORE ENTITIES INCLUDE:

*USER • ARTICLE • CATEGORY • SOURCE • FAVORITE • READINGHISTORY
• BOOKMARK • NOTIFICATION • FEEDBACK • ADMINACTION • TAG •
COMMENT*

RELATIONSHIPS BETWEEN THESE ENTITIES SUPPORT THE PLATFORM'S KEY OPERATIONS: A USER RELATES TO MANY FAVORITES, BOOKMARKS, AND READING HISTORY RECORDS; EACH ARTICLE BELONGS TO A CATEGORY AND SOURCE AND CAN CARRY MULTIPLE TAGS AND COMMENTS; AND NOTIFICATIONS AND FEEDBACK ARE LINKED BACK TO THE USER WHO TRIGGERED THEM. APPROPRIATE INDEXES AND FOREIGN-KEY CONSTRAINTS ENSURE EFFICIENT LOOKUPS FOR PERSONALIZED FEED GENERATION AND TRENDING COMPUTATION.

3.1 PROJECT TIMELINE SUMMARY

THE WORK BREAKDOWN STRUCTURE DECOMPOSES THE PROJECT INTO NINE PHASES ACROSS A 12-WEEK SCHEDULE. SEVERAL PHASES (CORE BACKEND, AI AGENT, AND FRONTEND) ARE EXECUTED IN PARALLEL FROM WEEK 5 ONWARD ONCE THE DATABASE SCHEMA AND SYSTEM ARCHITECTURE ARE STABLE, COMPRESSING THE SEQUENTIAL EFFORT BELOW INTO THE 12-WEEK DELIVERY WINDOW.

SECTION-CACHING, RESPONSE STORAGE & VECTOR DATABASE

BECAUSE AI SUMMARIZATION AND CLASSIFICATION CALLS ARE THE MOST EXPENSIVE OPERATIONS IN THE SYSTEM, THE DESIGN INCLUDES AN EXPLICIT CACHING AND RESPONSE-STORAGE STRATEGY RATHER THAN TREATING THE LLM AS A STATELESS CALL MADE ON EVERY REQUEST:

- **CACHING** - FREQUENTLY ACCESSED DATA — TRENDING TOPIC LISTS, CATEGORY-LEVEL ARTICLE COUNTS, AND HOT ARTICLE METADATA — IS CACHED IN AN IN-MEMORY STORE (E.G. REDIS) WITH SHORT, TIME-BOXED EXPIRY, REDUCING REPEATED LOAD ON THE RELATIONAL DATABASE FOR READ-HEAVY ENDPOINTS SUCH AS THE HOME FEED AND TRENDING PAGE.
- **AI RESPONSE STORAGE** : OUTPUTS FROM THE LOCAL LLAMA 3.2 AGENT (SUMMARIES, CLASSIFICATIONS, TRANSLATIONS) ARE PERSISTED BACK INTO THE RELATIONAL DATABASE AGAINST THE ORIGINATING ARTICLE, KEYED BY ARTICLE ID AND OPERATION TYPE. ONCE AN ARTICLE HAS BEEN SUMMARIZED OR CLASSIFIED, SUBSEQUENT REQUESTS FOR THE SAME ARTICLE AND OPERATION ARE SERVED FROM STORAGE INSTEAD OF RE-INVOKING THE MODEL, WHICH KEEPS RESPONSE LATENCY LOW AND AVOIDS REDUNDANT LOCAL INFERENCE.
- **VECTOR DATABASE**: NOT USED IN THE CURRENT PHASE OF THE PROJECT. SIMILARITY SEARCH OVER ARTICLE CONTENT (E.G. FOR “RELATED ARTICLES” OR SEMANTIC SEARCH) IS A CANDIDATE EXTENSION IDENTIFIED FOR A LATER PHASE; IF ADOPTED, ARTICLE EMBEDDINGS WOULD BE STORED IN A DEDICATED VECTOR STORE (E.G. PGVECTOR OR A STANDALONE VECTOR DATABASE) RATHER THAN THE CORE

RELATIONAL SCHEMA, KEEPING TRANSACTIONAL AND SIMILARITY-SEARCH CONCERNS SEPARATE.

THIS CACHING AND RESPONSE-STORAGE LAYER IS WHAT ALLOWS THE PLATFORM TO OFFER AI-DRIVEN FEATURES WITHOUT EVERY PAGE LOAD TRIGGERING A FRESH MODEL INFERENCE, AND IT IS TREATED AS A FIRST-CLASS PART OF THE DATABASE DESIGN RATHER THAN AN INCIDENTAL IMPLEMENTATION DETAIL.

SECTION -> TABLE OF DURATION -

<u>PHASE</u>	<u>EST. DURATION</u>
1. INITIATION	2 WK
2. REQUIREMENTS & DESIGN	3.5 WK
3. DATABASE DESIGN	2 WK
4. CORE BACKEND	6 WK
5. AI AGENT	4 WK
6. FRONTEND	4.5 WK
7. FEATURE MODULES	5 WK
8. INTEGRATION & TESTING	4 WK
9. DEPLOYMENT & CLOSURE	2.5 WK

4. ENTITY-RELATIONSHIP (ER) DIAGRAM

THE ER DIAGRAM FOR THE ENTITY RELATIONAL SCHEMA DESCRIBED IN SECTION 3 IS INSERTED BELOW.

